

Peijin Sun Ph.D.

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Google Scholar: [Peijin Sun](#) ORCID: <https://orcid.org/0000-0001-6054-2950> No visa sponsorship required.

Senior Geospatial Data Analyst with 10+ years of experience in urban planning, geospatial analytics, and data science. Expert in leveraging large-scale multimodal data and AI/ML to drive regional decision-making and improve quality of life. Proven track record in translating complex datasets into data stories and policy recommendations for governmental and community stakeholders.

Technical Skills

- **Geospatial Data Analytics:** Python (Pandas, GeoPandas) | ArcGIS Suite | SQL | Git | Spatial Statistics | Environmental Exposure Assessment | Real-World Data Analytics
- **Data Science & Computational Workflows:** Statistical Modeling | Causal Inference | Machine Learning | Predictive Modeling | Multimodal Data Integration | Agentic AI Workflows
- **Research:** Behavioral & Health Analytics | Mixed-Methods Research | Real-World Evidence Studies
- **Communication & Reporting:** Data Visualization | Interactive Dashboards | Data Storytelling for non-technical audiences | Cross-Functional Stakeholder Communication | Adobe Creative Suite

Work Experience

AI Product Manager (Health Data Platform), Mengyan Intelligence Ltd. Jan 2026 – Present

- Leading the design of an Agentic AI-driven health analytics platform integrating clinical records, behavioral data, and environmental exposures to support real-world studies and decision-support tools.
- Translate research questions into product requirements, AI model workflows, and member-facing analytics concepts, partnering with cross-functional teams to deliver scalable digital health solutions.

Postdoc, University of North Carolina at Chapel Hill, Chapel Hill, NC Sep 2024 – Dec 2025

Assistant Professor, Dalian University of Technology, Dalian, China Jul 2020 – Sep 2024

- **Research & Project Management:** Environmental health research translating complex problems into analytical frameworks. 20+ journal articles, 1 monograph; PI on 6 funded projects.
- **Large-Scale Multimodal Analysis:** Applied advanced spatial-statistical techniques to analyze large-scale structured and unstructured datasets (e.g., text, imagery, mobility data, geospatial data, survey data, demographic data, behavioral health data) to support data-driven decision-making.
- **Data Science & Communication:** Applied AI/ML methods for data mining, cleaning, analysis, and visualization, ensuring data integrity and statistical rigor. Translated complex analytical results into clear, actionable insights for diverse stakeholders; presented at 7+ conferences and taught 8+ courses.

Strategic Planning Consultant, Part-time, Dalian Urban Development Co., Ltd. Mar 2024 – Dec 2024

- Performed complex spatial and market feasibility assessments for medical mall developments, integrating demographic trends and healthcare access metrics to identify optimal investment sites.
- Combined heterogeneous market datasets with GIS evidence to produce actionable strategic reports, successfully guiding senior decision-makers on regional healthcare infrastructure positioning.

Geospatial Consultant, Part-time, Dalian Territorial Spatial Planning Co., Ltd. Mar 2021 – Jun 2024

- **Regional Master Planning:** Led geospatial data analysis for one comprehensive master plan and two specialized regional plans (coastal zone and green space), designing thematic maps and spatial reports to drive long-term land-use strategies.
- **Large-Scale Data Modeling:** Processed and modeled large-scale spatial datasets to provide inferential evidence for regional planning policies, ensuring all recommendations were rooted in rigorous geospatial analytics.
- **Digital Twin & Regional Coordination:** Collaborated with inter-governmental partners to develop urban data platforms and digital twin prototypes, enhancing the agency's capacity to monitor regional growth patterns and communicate complex spatial findings to support informed decision-making.

Education

Ph.D., Urban Planning		2020
<i>University of North Carolina at Chapel Hill / Dalian University of Technology (Joint Program)</i>		
Master of Engineering, Urban Systems Engineering	Dalian University of Technology	2016
Bachelor of Engineering, Urban Systems Engineering	Dalian University of Technology	2014

Books

Sun Peijin, Lu Wei. ***Building Healthy Communities: Environmental Survey and Analysis Method***. 2023.
Published in Chinese by China Architecture & Building Press.

Interdisciplinary Research Methods

- Integrates social survey methods with technologies such as big data and AI.
- Combines qualitative and quantitative approaches from geography, sociology, statistics, and computer science to study behavior in real-world contexts.

Data Processing, Modeling, and Evidence Translation

- Details the processes of data collection, cleaning, and integration, with a focus on commonly used statistical models such as factor analysis, mediation effects, structural equation modeling, and causal inference.

Research Projects (as Principal Investigator)

- [1] Multimodal Modeling of Coastal Environment and Visitor Behavior | 2024–2025
- [2] AI-Enabled Decision Support Using Multimodal Behavioral Data | 2024–2025
- [3] Behavioral–Spatial Data Integration for Rural Tourism Innovation | 2023–2025
- [4] Spatial Technologies and Behavioral Insights for Rural Revitalization | 2024
- [5] Smart City Curriculum Innovation through Emerging Technologies and Data Science | 2023–2024
- [6] Multi-Source Data Integration for Environmental Exposure and Physical Activity Analysis | 2020–2023

Publications

- [1] Sun, P., Liu, P. (2026). **Social Equity in Park Facilities: Assessing Park Usage Patterns Before and After the COVID-19 Pandemic**. *Urban Forestry & Urban Greening*, 117, 129249. (SCI, JCR Q1)
- [2] Sun, P., Zhao, H., Zhong, J., et al. (2025). **Popularity influence mechanism of coastal spaces in urban areas: Insights from multi-modal large language models**. *Cities*, 161, 105909. (SCI, JCR Q1)
- [3] Sun, P., Sun, J., Jin, L., & Zhu, Y. (2024). **COVID-19 pandemic changes the outdoor physical activity preference in Chinese city: A 7-year GPS trajectory data analysis**. *Cities*, 152, 105253. (SCI, JCR Q1)
- [4] Sun, P., Zhao, H., & Lu, W. (2024). **How urban environments affect public sentiment and physical activity using a cognitive computing framework**. *Frontiers of Architectural Research*, *Frontiers of architectural research*, 2024-01. (A&HCI)
- [5] Sun, P., Liu, P., & Song, Y. (2024). **Seasonal variations in urban park characteristics and visitation patterns in Atlanta: A big data study using smartphone user mobility**. *Urban Forestry & Urban Greening*, 91, 128166. (SCI, JCR Q1)
- [6] Sun, P., Lu, W., & Jin, L. (2023). **How the natural environment in downtown neighborhood affects physical activity and sentiment: Using social media data and machine learning**. *Health & Place*, 79, 102968. (SCI, JCR Q1)
- [7] Sun, P., Song, Y., & Lu, W. (2022). **Effect of Urban Green Space in the Hilly Environment on Physical Activity and Health Outcomes: Mediation Analysis on Multiple Greenery Measures**. *Land*, 11(5). (SSCI, JCR Q1)
- [8] Sun, P., & Lu, W. (2022). **Environmental inequity in hilly neighborhood using multi-source data from a health promotion view**. *Environmental Research*, 204(Pt A), 111983. (SCI, JCR Q1)

- [9] Wan, T., Lu, W., & **Sun, P.** (2023). *Equity impacts of the built environment in urban rail transit station areas from a transit-oriented development perspective: a systematic review*. Environmental Research Communications, 5(9), 92001. (SCI, JCR Q2)
- [10] Jin, L., Lu, W., & **Sun, P.** (2022). *Preference for Street Environment Based on Route Choice Behavior While Walking*. Frontiers in Public Health, 10, 880251. (SCI, JCR Q1)
- [11] Jin, L., Lu, W., & **Sun, P.** (2022). *Effect of the Street Environment on Walking Behavior: A Case Study Using the Route Choice Model in the Chunliu Community of Dalian*. Frontiers in Public Health 10(May):874788. (SCI, JCR Q1)
- [12] Wan, T., Lu, W., & **Sun, P.** (2022). *Constructing the Quality Measurement Model of Street Space and Its Application in the Old Town in Wuhan*. Frontiers in Public Health 10(February):1–18. (SCI, JCR Q1)
- [13] Sun, Y., Lu, W., & **Sun, P.** (2021). *Optimization of Walk Score Based on Street Greening—a Case Study of Zhongshan Road in Qingdao*. International Journal of Environmental Research and Public Health 18(3):1–13. (SCI, JCR Q1)
- [14] **Sun, P.**, Song, Y., Lu, W., & Gu, Z. (2019). *Influences of Built Environment with Hilly Terrain on Physical Activity in Dalian, China: An Analysis of Mediation by Perceptions and Moderation by Social Environment*. International Journal of Environmental Research and Public Health 16(24):1–17. (SCI, JCR Q1)

Publications in Chinese Journal

- [1] **Sun Peijin**, Lu Wei, Wu Liang. *Differences of Health-Oriented Built Environment Assessment Methods and Scales*, Modern Urban Research, 2020, 35(4):36-43.
- [2] **Sun Peijin**, Lu Wei. *Institutional-Oriented Design for Health and Its Enlightenment to China: "Active Living by Design"* International Progress and Implications, Journal of Human Settlements in West China, 2020, 2:60-66.
- [3] **Sun Peijin**, Lu Wei, Liu Lianlian. *Design Guidelines for Active Living: Western Experience*, Urban Planning International, 2019, 6:170-178.
- [4] **Sun Peijin**, Lu Wei. *The Correlation between Urban Green Space and Residents' Physical Activity and Health Outcome*, South Architecture. 2019, 3:34-39.
- [5] **Sun Peijin**, Song Yan, Lu Wei. *Healthy Planning Implementation System and Policy: Learning from International Experience of the United States*, Shanghai Urban Planning Review, 2019, 5:117-122.
- [6] **Sun Peijin**, Lu Wei, Wu Liang, *The Association between Urban Design Quality and Walking Behavior in Dalian*, New Architecture, 2019, (05):97-101.
- [7] Wan Tianyue, Lu Wei, Sun Peijin. *Progress and Prospect of Spatial Equity in Public Service Facilities Based on Bibliometric*, Urban & Rural Planning, 2023, (05), 93-105
- [8] Jin Lan, Lu Wei, Sun Peijin. *Walkability Measurement of the Built Environment and Its Validity in Dalian*, New Architecture, 2022, (04):100-105.
- [9] Xin Yuzheng, Lu Wei, Sun Peijin. *Research and Prospect of Green Space from the Perspective of Public Mental Health*, Landscape Architecture, 2022, 29(03):79-85.

Honors and Awards

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| • Distinguished Science and Technology Talent of Dalian (Awarded RMB 300,000) | 2021 |
| • Outstanding Doctoral Dissertation Award , Dalian University of Technology | 2021 |
| • Excellent Graduate Student , Dalian University of Technology | 2020 |
| • CSC Scholarship Program , China Scholarship Council | 2017 |
| • Excellent Undergraduate , Dalian University of Technology | 2014 |

Conference Oral Presentations

- [1] Peijin Sun. Mechanisms influencing the popularity of multimodal data-driven coastal Spaces: application of large language models, China Tourism Geography Academic Conference, South China University of Technology, Guangzhou, 2024, 7.6-7.

- [2] Peijin Sun, Daiyun Liu. Exploration and Teaching Practices in Coastal Space Planning and Design in Dalian, Special Committee on Master Planning, Urban Planning Society of China, Dalian University of Technology, Dalian, 2024, 6.21-23.
- [3] Peijin Sun. How the natural environment in downtown neighborhoods affects physical activity and sentiment: Using social media data and machine learning, Health Geography Youth Forum, Beijing Normal University, Zhuhai, 2024, 5.23-24.
- [4] Peijin Sun. How urban environments affect public sentiment and physical activity using a cognitive computing framework, Chinese Healthy City Science Annual Conference, Tongji University, Shanghai, 2023, 10.26-27.
- [5] Peijin Sun. The Application of Language Models in the Study of Urban Environments and Residents' Physical Activity, The 17th International Association for Chinese Planning, Tianjin University, Tianjin, 2023, 6.28-7.02.
- [6] Peijin Sun. Relationship between urban environment and physical activity based on social media data, The 1st Environment and Behavior International Symposium, Online, 2022, 11.19-20.
- [7] Peijin Sun. Environmental inequity in hilly neighborhood using multi-source data from a health promotion view, Sino-UK Joint Symposium on Post Novel Covid-19 Pandemic, Online, 2021, 8.27-29.

Teaching Innovation

Undergraduate courses: Urban Digital Technology; Spatial Quantitative Analysis; Urban Cognition Survey; Urban Park Planning and Design; Coastal Landscape; Design Basis

Graduate courses: Urban Problem Survey; Urban Planning and Design; Academic Writing.

Innovative Courses Developed on Smart Cities

- **Urban Digital Technology**

Key Topics: Smart Cities, Digital Twins, Geographic Information Systems, Artificial Intelligence, Machine Learning, Computer Vision, Natural Language Processing, Internet of Things, Sensors, Data Visualization.

- **Spatial Quantitative Analysis/ Urban Problem Survey**

Focus: A practical course designed to apply advanced technologies, including machine learning and spatial analysis, to tackle complex urban challenges.

Application: Guided students through hands-on projects applying data-driven and design-thinking approaches to solve real-world challenges.

Co-Founder | AIMorph Lab

An interdisciplinary design studio that applies advanced mathematical modeling to develop tangible products through parameterized design and 3D printing. Successfully launched a consumer-ready product line now available on e-commerce platforms.

- Led end-to-end project execution for a startup venture, coordinating cross-functional tasks from concept design and prototyping to supply chain alignment and vendor management.
- Conducted data-driven market research and user analysis to inform product strategy and positioning; translated insights into branding decisions and product-market fit optimization.



$$x_{n+1} = x_n^2 - y_n^2 + ax_n + by_n$$

$$y_{n+1} = 2x_n y_n + cx_n + dy_n$$
 Inspired by the Tinkerbell Map - a discrete-time dynamical system - this ring design captures its spiraling curves.

Life stretches wide like the universe, quietly holding infinite wonder and delight. Even the simplest choice may lead us to unexpected beauty. May we shine like the stars - curious, courageous, and ever-dreaming.
